

Optimization of an Airfoil Characteristics to Minimize the Turn Radius of a Small Unmanned Aerial Vehicle

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Abstract—Recently, one of the priorities of the general aviation all around the world is to reduce the environmental hazards of aviation. Optimization is the endpoint to be done in order to diminish the difference between energy used and energy supplied for aircraft. These ideas bring some new methods and tools to our mind for aerodynamic shape optimization should be developed and allowing access to aircraft configuration that could not be obtained with ancient techniques. This paper has mainly focused on an application of optimization tool that can reach the advanced aircraft design parameters. This research targets to optimizing an airfoil by changing the parameters such as maximum thickness, camber, camber location, the angle of attack, and the wing taper ratio in order to minimize the turn radius of a small unmanned aerial vehicle at low Reynolds numbers. Firstly, the base airfoil, which is NACA4412, is generated by using MATLAB [1] and it is analyzed in XFOIL. Then, the aerodynamic characteristics of the airfoil are evaluated thanks to MATLAB-XFOIL interface. Eventually, the airfoil is optimized by using MATLAB fmincon solvers, which are Sequential Quadratic Programming and Interior-Point Methods. The results - coming from solvers - are compared. During the iterative process, the lift coefficient and the moment coefficient of the airfoil are kept constant. The drag coefficient of the airfoil and the induced drag factor are minimized to minimize the turn radius. This work in this paper may have a certain significance for designing an optimum airfoil in order to minimize the turn radius of an unmanned aerial vehicle.

Keywords—airfoil; thickness; camber; camber location; taper ratio; tightest turn; MATLAB; fmincon; XFOIL

I. INTRODUCTION

Turn is an essential part of the flight maneuvers. In a turn motion, the aircraft flies through all or part of a circle. This circle or loop could be horizontal, vertical, or with an angle relative to the horizontal. In air war, the defeater is generally the aircraft that has the smallest turning radius or the highest turn rate. In this paper, the level turning flight will be considered. It is well known that the level flight requires the forces in the vertical direction to be balanced. It is clearly seen that this requirement limits the turning performance. One of the needs is the structural limit and the other one is the aerodynamic limit. This study focuses on the aerodynamic limit, and it will be forced to reach the

optimum design condition to get the minimum turn radius by optimizing airfoil shape.

Airfoil design for unmanned aerial vehicles is an essential part of UAV development process. The airfoils currently used are pretty old like the NACA family, which was originally developed for airplanes during World War II. However, today, in order to obtain an optimum airfoil shape satisfying the mentioned requirements, several techniques have been applied during the designing sequences, through numerical optimization [2].

The objective of the present work is to automatically generate airfoil profiles in MATLAB by using the analytical formula and evaluate their aerodynamic performances via XFOIL, which is an interactive program for the design and analysis of subsonic isolated airfoils. After having various profiles with variable performance the optimum one among which has been selected by using MATLAB fmincon function, which is one of the numerical optimization tools. fmincon attempts to find a constrained minimum of a scalar function of several variables starting at an initial estimate and it uses several solvers such as Sequential Quadratic Programming, Active Set, and Interior Point. In this study, the problem is solved by using Sequential Quadratic Programming and Interior Point Methods the results obtained are compared.

II. METHODOLOGY

A. Airfoil Generation

The airfoils can be represented by type functions or spline curves. Even though the obtained results are good enough, more than 20 variables are needed in order to represent the airfoil accurately but the calculations take a long time. By using analytical functions to define the airfoil, such as NACA series, the calculation time can be reduced and the airfoil can be defined by only 5 variables [3].

The four digits in NACA, these 4 digits in airfoils designate the camber (m), maximum camber location (p), and the thickness (t) which all are expressed in terms of the ratio of wing root to chord (c_r).

The thickness distribution (y_t) an equation for NACA 4 digit airfoil,

$$y_t = \frac{t}{0.2} (a_0 x^{0.5} + a_1 x + a_2 x^2 + C) \quad (1)$$

Where

$$C = a_3x^3 + a_4x^4 \quad (2)$$

and a_0, a_1, a_3 , and a_4 are constants and equal to 0.2969, -0.126, -0.3516, and -0.1036 for a closed trailing edge, respectively.

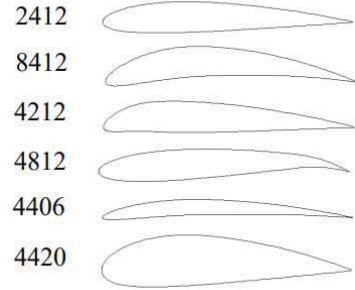
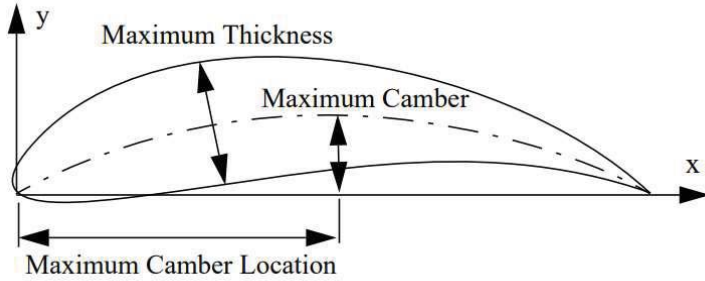


Figure 1. Naca 4 digit airfoil parameters. [4]

The analytical function defining the airfoil camber for the x values of between 0 and (p),

$$y_c = \frac{m}{p^2}(2px - x^2), \quad (0 \leq x < p) \quad (3)$$

and for the x values of between (p) and 1

$$y_c = \frac{m}{(1-p)^2}(1 - 2p - 2px - x^2) \quad (4)$$

(x) is the coordinate in airfoil chord direction, from 0 to (c) and (y_c) represents the airfoil camber coordinate.

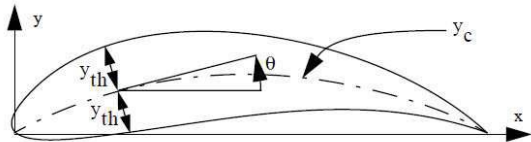


Figure 2. Camber line angle. [4]

The gradient equations are given for the x values of between 0 and (p), and the x values of between (p) and (c) in equations (5) and (6), respectively.

$$\frac{dy_c}{dx} = \frac{2m}{p^2}(p - x) \quad (5)$$

$$\frac{dy_c}{dx} = \frac{2m}{(1-p)^2}(p - x) \quad (6)$$

The airfoil upper surface coordinates (x_u, y_u) and lower surface coordinates (x_l, y_l) can be defined as;

$$x_u = x_c - y_t \sin \theta \quad (7)$$

$$y_u = y_c + y_t \cos \theta \quad (8)$$

$$x_l = x_c + y_t \sin \theta \quad (9)$$

$$y_l = y_c - y_t \cos \theta \quad (10)$$

Where

$$\theta = \tan^{-1} \frac{dy_c}{dx} \quad (11)$$

By substituting equations (1), (3), and (4) into equations (7), (8), (9), and (10), for x values between 0 and (p), $0 \leq x < p$, the lower and the upper coordinates are described as;

$$x_u = x - 5t(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \sin \theta \quad (12)$$

$$y_u = \frac{m}{p^2}(2px - x^2) + 5t(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \cos \theta \quad (13)$$

$$x_l = x + 5t(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \sin \theta \quad (14)$$

$$y_l = \frac{m}{p^2}(2px - x^2) - 5t(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \cos \theta \quad (15)$$

For x values between (p) and (c), $p \leq x \leq c$, the lower and the upper coordinates are described as;

$$x_u = x - 5t(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \sin \theta \quad (16)$$

$$y_u = \frac{m}{(1-p)^2}((1 - 2p) + 2px - x^2) + 5t(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \cos \theta \quad (17)$$

$$x_l = x + 5t(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \sin \theta \quad (18)$$

$$y_l = \frac{m}{(1-p)^2} ((1-2p) + 2px - x^2) - 5t(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4) \cos \theta \quad (19)$$

The equations given above represent the airfoil border lines as expressed. One of them is the upper boundary and the other one is the lower boundary of the airfoil [5].

B. Optimization Function, Design Variables and Constraints

The aim of this paper is to obtain the possible minimum turn radius of a small unmanned aerial vehicle as stated before. The turn radius (R) is expressed as [5]

$$R = \frac{V_\infty^2}{g\sqrt{n^2-1}} \quad (20)$$

In (20), gravitational acceleration (g) is constant. Hence, the turn radius depends on only the velocity (V_∞) and the load factor (n).

The minimum value of (R) is obtained when the derivative of the turn radius with respect to velocity is equal to zero, $dR/dV_\infty = 0$. The algebra will be simplified if the equation (20) is written in terms of the dynamic pressure (q_∞).

$$q_\infty = \frac{1}{2} \rho_\infty V_\infty^2 \quad (21)$$

Equation (20) becomes;

$$R = \frac{2q_\infty}{g\rho_\infty\sqrt{n^2-1}} \quad (22)$$

By differentiating equation (22) with respect to q_∞ and setting it equal to zero, the following equation (23) is obtained. It is important to notice that the load factor is a function of V_∞ hence q_∞ .

$$\frac{dR}{dq_\infty} = \frac{2A\sqrt{B}-2Aq_\infty n B^{-1/2} dn/dq_\infty}{A^2 B} = 0 \quad (23)$$

where

$$A = g\rho_\infty \quad (24)$$

and

$$B = n^2 - 1 \quad (25)$$

Equation (23) can be rewritten as;

$$n^2 - 1 - q_\infty n \frac{dn}{dq_\infty} = 0 \quad (26)$$

The load factor can be given as a function of dynamic pressure [3]

$$n^2 = \frac{q_\infty}{K(W/S)} \left(\frac{T}{W} - q_\infty \frac{C_{D,0}}{W/S} \right) \quad (27)$$

Differentiating equation (27) with respect to q_∞ gives

$$n \frac{dn}{dq_\infty} = \frac{T/W}{2K(W/S)} - \frac{q_\infty C_{D,0}}{K(W/S)^2} \quad (28)$$

If equation (26) is rewritten by substituting equations (27) and (28)

$$q_\infty = \frac{2K(W/S)}{T/W} \quad (29)$$

Hence, the velocity corresponding to the minimum turning radius ($(V_\infty)_{R_{min}}$)

$$(V_\infty)_{R_{min}} = \sqrt{\frac{4K(W/S)}{\rho_\infty(T/W)}} \quad (30)$$

By using the definition of the load factor (30) becomes

$$n_{R_{min}} = \sqrt{2 - \frac{4KC_{D,0}}{(T/W)^2}} \quad (31)$$

The expression for minimum turning radius is obtained by substituting equations (30) and (31) into equation (20);

$$R_{min} = \frac{4K(W/S)}{g\rho_\infty(T/W)\sqrt{1-4KC_{D,0}/(T/W)^2}} \quad (32)$$

The wing loading (W/S) is constant and equal to 30.78125 kg/m^2 . Thrust-to-weight ratio (T/W) is also constant and has the value of 1.11. These values are not change for the examined UAV at certain a condition. Hence, the only coefficients that affect the minimum turn radius are induced drag factor (K) and the zero-lift drag coefficient ($C_{D,0}$).

The induced drag factor (K) simply depends on Oswald span efficiency (e) and the aspect ratio (AR) [5].

$$K = \frac{1}{\pi e AR} \quad (33)$$

The aspect ratio of the aircraft is defined as;

$$AR = \frac{b^2}{S} \quad (34)$$

and the Oswald span efficiency (e)

$$e = \frac{1}{1+d} \quad (35)$$

(d) is calculated by lifting line theory and it is a function of taper ratio and the aspect ratio¹ as seen from Figure 3[5]. The aspect ratio is equal to 10 as mentioned before. For the values of taper ratio between 0 and 1, (d) values are found by digitizing Figure 3 by using GetData Graph Digitizer, which is a program to digitize graphs and plots, and the results are indicated in TABLE I. And using equations (33) and (35), (K) is obtained.

The parasite drag ($C_{D,0}$) is taken as equal to the drag coefficient of the airfoil (c_d) at zero lift.

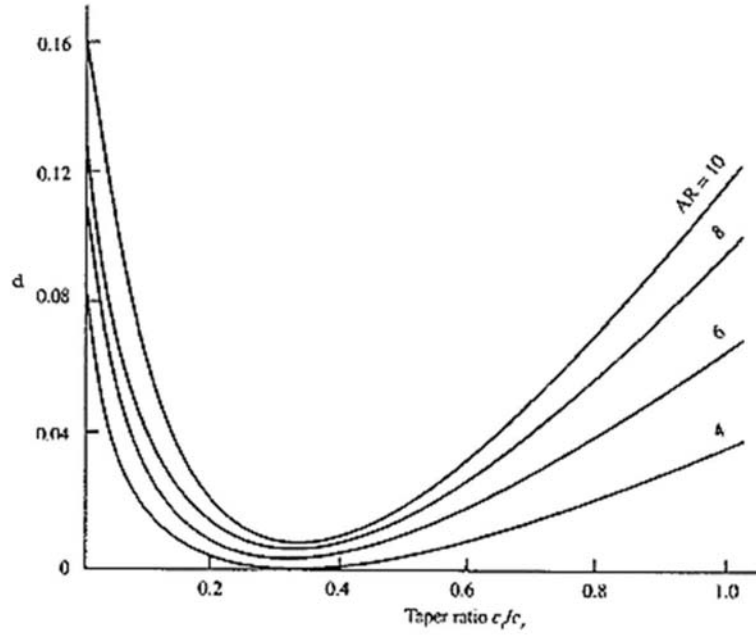


Figure 3. Induced drag factor as a function of taper ratio for wings of different aspect ratios[5]

TABLE I. d VALUES AT DIFFERENT TAPER RATIOS FOR AR OF 10

Taper ratio	d
0.1	0.058
0.2	0.0195
0.3	0.0087
0.4	0.00111
0.5	0.0019
0.6	0.035
0.7	0.054
0.8	0.074
0.9	0.0945
1.0	0.117

NACA 4-digit airfoils are determined by variables camber (m), maximum camber location (p), and thickness (t) as seen from airfoil generation part. Therefore, NACA4412 is chosen as the basic airfoil and (m), (p), (t), the wing taper ratio (λ), and the angle of attack (α) are selected as design variables. Initial values are taken as 0.04, 0.4, 0.12, 0.9, and -2.625° , respectively.

To describe the general optimization problem R_{min} can be defined as F as follows;

$$\min F = \frac{4K(W/S)}{g\rho_\infty(T/W)\sqrt{1-4KC_{D,0}/(T/W)^2}} \quad (36)$$

Subject to

$$c_m + 0.0796 = 0$$

$$c_l - 0.078 \leq 0$$

$$0 \leq m \leq 0.1$$

$$0 \leq p \leq 0.8$$

$$0.072 \leq t \leq 0.18$$

$$0 \leq \lambda \leq 1$$

$$-5^\circ \leq \alpha \leq 15^\circ$$

where (c_m) is the moment coefficient of the airfoil and (c_l) is the lift coefficient of the airfoil.

C. Optimization Tool

1) *fmincon*

The optimization problem is solved by using MATLAB-XFOIL interface. Firstly, the airfoil is generated via MATLAB. Then, the airfoil is analyzed by using XFOIL at Reynolds number of 120,000 and Mach number of 0.0588. After the analysis is completed, the drag coefficient of the airfoil (c_d) value is taken from XFOIL and it is substituted into the objective function in equation (38). The airfoil maximum thickness, camber, the camber location, and the wing taper ratio are kept as variables if the possible minimum value of the objective function is not obtained. The process continues until the possible minimum value of the objective function is obtained. During the process, the lift coefficient of the airfoil (c_l) and the moment coefficient of the airfoil (c_m) are kept constant as mentioned before.

Solvers are generally found a local solution satisfying the Karush-Kuhn-Tucker (KKT) optimality conditions. MATLAB *fmincon* is used to solve the minimization problem by choosing the Interior-Point Method and the Sequential Quadratic Programming Method as the solvers. The interior point methods, also called as barrier functions, are linear or nonlinear programming methods. In Interior Point method because of their “combinatorial” nature inequality constraints are deceptive to overcome. It can be said either g_k is active or its associated dual variable is 0. This can be explained by the KKT complementarity condition [6]:

$$\mu_k g_k(x) = 0 \text{ for each } k$$

Interior Point Method does not address an above problem directly, it works on its relaxation as $\mu_k g_k(x) = v$ with $v > 0$.

Sequential Quadratic Programming methods refine iteratively estimations which were estimated that which

inequality constraints g_k are active. At the current iteration, on-binding constraints have no influence. That's why the solver then works on the null space of remaining constraints. [6]

TABLE II. COMPARISON OF SQPM AND IPM [7]

Sequential Quadratic Programming Method	Interior Point Method
• <i>Initialization</i>	<i>Initialization</i>
Structure of a quadratic model of the problem	• There is no initialization procedure
Try a first estimation of the set of active constraints	
• <i>Each Iteration</i>	• <i>Each Iteration</i>
Compute the quadratic model, start it by the binding set estimation	Compute one Newton iteration towards the solution of the relaxed KKT system
Updates new variable(x_{k+1}) and the estimation of the active constraint	Updates the current solution estimate and the relaxation parameter v
Structure a new quadratic model at new variable (x_{k+1})	
• <i>Properties</i>	• <i>Properties</i>
Performs lots of computational inexpensive iterations	Perform some computationally expensive iterations
Work on the null space of the active constraints	Rely on efficient underlying linear algebra

All in all, it can be said that Interior Point Method solvers scale very well to large Non-Linear Programming problems with a **small number of constraints** although the Sequential Quadratic Programming solvers scale very well to large Non-Linear Programming problems with a **high number of constraints** [7].

2) Genetic algorithm

For the second optimization section a single-objective multi-nonlinear and linear constraints genetic algorithm is applied with the help of MATLAB interface. In genetic algorithm it is used population-based method like there is a gene-pool that continuously regenerated and refreshed with new generation [8]. There are two parents and randomly they are selected. Offspring is created from them and it is multiplied with cross-over and mutation operator. After that the performance of the new offspring is evaluated with the help of XFOIL. At the end of the iteration there are N survivals and they are the best gene-type.

III. RESULTS AND DISCUSSION

The objective function has reached the local minimum value at the eleventh iteration and the process lasted 352.257 seconds when solved with Interior-Point method. When solved with SQP method, the optimum value is obtained at fourteenth iteration and it takes 470.414 seconds. The constraints are satisfied in both methods. It is seen that accurate values are obtained in a very short time. Also the genetic algorithm is used to optimize given airfoil and it takes 780.789 seconds.

The initial and optimized values of the design variables are demonstrated in TABLE III. Also the initial and optimized values of the parasite drag shown in Table IV with the turn radius.

The velocity at minimum turn radius (V_{Rmin}) in Interior-Point Method and SQP method are found as 6.2670 m/s and 6.2642 m/s, respectively by using equation (30). The induced factor (K) corresponding to optimized taper ratio is 0.0321 for both methods.

In Figure 4, the original NACA4412 airfoil and the optimized airfoils are presented. Also the iteration numbers and corresponding function values is given in Figure 5.

TABLE III. INITIAL AND OPTIMIZED VALUES OF THE DESIGN VARIABLES

	Initial value	IPM	SQPM	GA
Maximum camber	0.04	0.0522	0.0370	0.0539
Camber location	0.40	0.3376	0.623	0.3440
Thickness	0.12	0.0622	0.0652	0.0832
Taper ratio	0.90	0.5102	0.4671	0.5313
Angle of attack	2.625°	3.0°	2.285°	2.52°

TABLE IV. INITIAL AND OPTIMIZED VALUES OF PARASITE DRAG AND TURN RADIUS

	Initial val	IPM	SQPM	GA
Parasite drag	0.0220	0.0170	0.0172	0.0168
Turn R. (m)	3.5825	3.3845	3.3847	3.3692

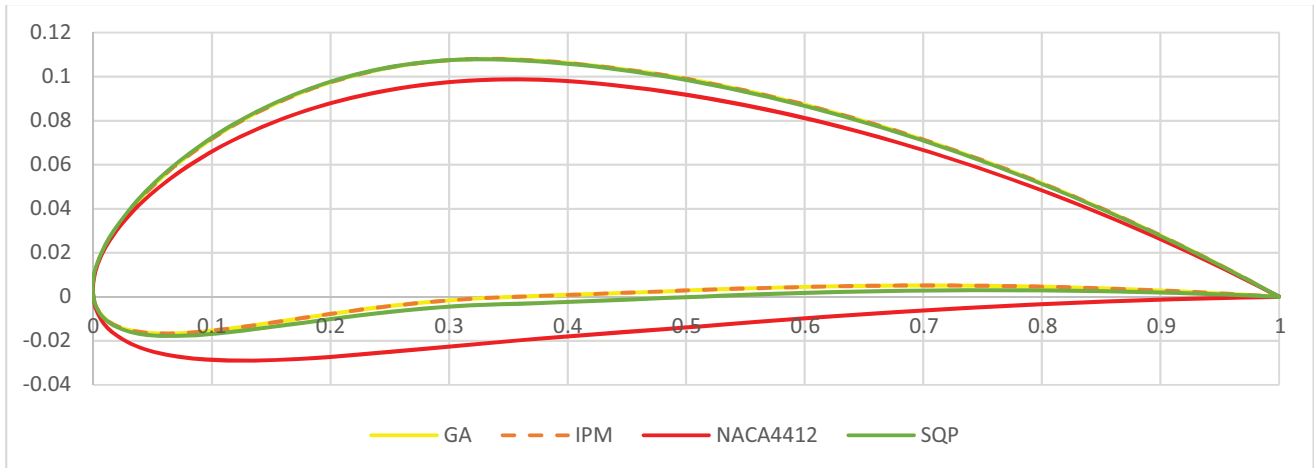


Figure 4. Original NACA4412 airfoil and the optimized airfoils.

After all these calculations, manipulations, and analysis what did we gain? It is clearly seen that there is a significant reduction at turn radius parameter. There are various application areas of this type of airfoil. Agricultural Spraying

UAV is one of them. It makes multiple turns while spraying a field. The minimum turn radius condition is the essential factor for this kind of UAV [9].

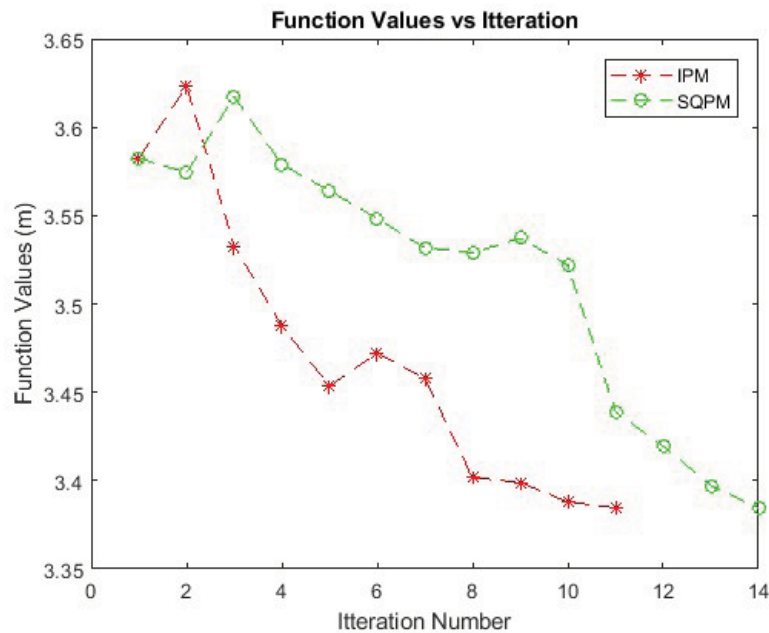


Figure 5. Function values

IV. CONCLUSION

In this paper, NACA4412 airfoil is optimized in order to minimize the turn radius of a small unmanned air vehicle. First of all, the airfoil is generated in MATLAB by using NACA 4-digit airfoil analytical formula. The airfoil analyzed by using XFOIL. The aerodynamic properties of the airfoil such as lift coefficient, moment coefficient, and the drag coefficient are evaluated by using MATLAB-XFOIL interface. The airfoil maximum thickness, camber, the camber location, the wing taper ratio, and angle of attack are changed until the turn radius is minimized. The optimization

problem is solved by MATLAB fmincon solvers, Sequential Quadratic Programming and Interior Point, and also with genetic algorithm. It is seen that used solvers give reasonable results when compared to historical data.

APPENDIX

AR	aspect ratio
b	wing span
c_d	drag coefficient of airfoil
$C_{D,0}$	parasite drag
c_l	lift coefficient of airfoil

c_m moment coefficient of airfoil
 c_r root chord
 c_t tip chord
 e Oswald span efficiency
 g gravitational acceleration
 n load factor
 $n_{R_{min}}$ load factor at minimum turn radius
 K induced drag factor
 m camber of airfoil
 p maximum camber location of airfoil
 R turn radius
 S wing area
 t thickness of airfoil
 T/W thrust to weight ratio
 UAV unmanned aerial vehicle
 V_∞ velocity of the air
 $(V_\infty)_{R_{min}}$ velocity at minimum turn radius
 q_∞ dynamic pressure
 W weight of the UAV
 W/S wing loading
 λ taper ratio
 ρ_∞ density

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